



Fig. 1 — Federated BIM Model — MEP Discipline Layers

REVIT MEP

REFERENCE GUIDE

- System Families
- Routing
- Coordination
- Schedules
- Clash Detection
- IFC Export

MEP
Systems

BIM
Coordination

Revit
Workflows

IFC
Export

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BIM & Revit MEP

Building Information Modelling Fundamentals

What is BIM?

Building Information Modelling (BIM) is a process for creating and managing digital representations of physical and functional characteristics of a facility. Revit MEP is Autodesk’s dedicated tool for mechanical, electrical, and plumbing engineering within the BIM environment, enabling parametric modelling, automated schedules, and multi-discipline coordination.



Revit MEP vs. AutoCAD MEP

Feature	Revit MEP	AutoCAD MEP
Model Type	Parametric 3D BIM	2D with 3D elements
Data Intelligence	Full MEP system data	Limited attributes
Schedule Generation	Automatic from model	Manual or semi-auto
Clash Detection	Via Navisworks link	Limited native
Coordination	Federated model	Xref-based
Learning Curve	Steep (3–6 months)	Moderate
Industry Adoption	Standard (ISO 19650)	Diminishing new projects



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Project Setup

Templates, Worksets, Worksharing & Linked Models

Project Template Best Practices

- Use discipline-specific MEP template (metric or imperial) as starting point
- Load only required system families — keep template lightweight
- Pre-configure view templates for each drawing type (plan, section, schematic)
- Set project units, annotation, and shared coordinates before any modelling
- Create discipline browser organisation (Mechanical, Electrical, Plumbing, Coordination)
- Define project phases: Existing, Demolition, New Construction

WORKSHARING SETUP ORDER:

1. Enable Worksharing (Collaborate tab → Worksets)
2. Create Worksets: Shared Levels, MECH-Ductwork, ELEC-Conduit, PLUMB-Pipe
3. Save to BIM360/ACC or network central file
4. Each user creates local copy (not the central file directly)
5. Synchronise with Central at key milestones — avoid long gaps

Linking Discipline Models

- Insert tab → Link Revit → select Architectural / Structural model
- Link method: Auto — By Shared Coordinates (requires coordinate setup)
- Copy/Monitor: Grids, Levels, Columns — update when architect revises
- Manage Links panel: Reload / Unload / Manage links visibility
- Do NOT explode linked models — maintain live link for coordination
- Visibility control: Linked model visibility per view via VV/VG Revit Links tab

Workset Name	Discipline	Contents	Owner
Shared Levels & Grids	All	Levels, grids, reference planes	BIM Manager
ARCH-Link	Architecture	Linked arch model	Read-only
MECH-Ductwork	Mechanical	HVAC duct systems	Mechanical Eng.
MECH-Piping	Mechanical	Chilled water, steam, refrig.	Mechanical Eng.
ELEC-Power	Electrical	Conduit, cable tray, switchgear	Electrical Eng.
ELEC-Lighting	Electrical	Luminaires, circuits, panels	Electrical Eng.
PLUMB-Pipe	Plumbing	Domestic, drainage, fire	Plumbing Eng.

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Mechanical Systems

HVAC Ductwork, Equipment & System Configuration

Creating Mechanical Systems in Revit

- Place Air Terminals (diffusers, grilles, registers) on ceiling/wall faces
- Place Mechanical Equipment (AHU, FCU, VAV) at correct elevation
- Create System: Select terminal → Create System → Supply Air / Return Air / Exhaust
- Select equipment as Air Handling Unit for the system
- Use Duct (DU) command to draw ductwork — Revit auto-routes between terminals
- Flex Duct: Use for terminal connections (max 1.5m length per SMACNA guideline)

Duct Sizing in Revit

Mechanical Settings → Duct Settings → Sizing Method:
 Equal Friction (Pa/m): Enter 0.8–1.2 Pa/m for comfort systems
 Velocity: Enter max velocity per section type
 Analyze tab → Duct Pressure Loss Report — export to schedule
 Size Ducts: Select system → Duct System tab → Size Ducts (auto-size)
 Manual override: Select duct segment → Properties → Width / Height

Mechanical Equipment & Connectors

Connector Type	Usage	Flow Direction	System Classification
Air Connector	Duct connections	Supply / Return / Exhaust	Ductwork
Pipe Connector	Chilled water, HW, Refrig.	Flow In / Out	Piping
Electrical Connector	Equipment power supply	—	Power
Face-based	Wall/ceiling mounted equip.	—	Family type specific
Work Plane-based	Standalone equipment	—	Floor-level placement

- Insulation: Select duct → Add Insulation (Modify|Ducts) → specify thickness/material
- Duct Lining: Select duct → Add Lining → reduces internal area (check sizing)
- Duct Fittings: Revit auto-places elbows, tees, reducers when routing; load family if missing

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Electrical Systems

Power Distribution, Lighting & Panel Scheduling

Electrical Model Components

- Distribution Equipment: Panels, switchboards, transformers, generators, UPS
- Power Devices: Outlets, switches, communication outlets, emergency devices
- Lighting Fixtures: Ceiling-hosted or face-based; load family with IES photometric data
- Conduit & Cable Tray: Route from equipment to devices; set conduit type and size
- Electrical Circuits: Connect devices to panel via Electrical → Power → Circuit

Panel Scheduling Workflow

1. Place Distribution Board (panel) family at correct location & voltage level
2. Set Panel Properties: Voltage, phases (1PH / 3PH), mains type, number of circuits
3. Create Circuits: Select devices → Modify tab → Power → Create Circuit → assign to panel
4. Panel Schedule: Analyse tab → Panel Schedules → Create/Modify → load count, VA per circuit
5. Verify load totals — Revit auto-calculates connected, estimated, demand loads
6. Export: Right-click panel schedule → Export (for client submission)

Conduit Routing

- Systems tab → Conduit with Fittings → Draw conduit run between devices and panels
- Conduit Type: Set EMT, RMC, PVC based on project spec (load via type properties)
- Conduit Size: Set manually (properties) or use Conduit Sizing (analyse tab)
- Cable Tray: Systems tab → Cable Tray → route backbone horizontal runs above ceiling
- Fittings: Revit auto-places standard elbows/tees; load manufacturer families for accuracy
- Conduit Stub-Up: Use Conduit Run that terminates at equipment connector point

Electrical System Type	Example Components	Revit Classification	Voltage Level
Normal Power (LV)	Lighting, general outlets	Power systems	230/400 V
Emergency/EL	Emergency lighting, fire panels	Power systems	230/400 V
Fire Alarm	FA devices, panels, loop	Data systems	24 VDC / 230 VAC
Data/BMS	IT outlets, BMS sensors	Communication	24 VDC / POE
MV Distribution	HV switchgear, transformers	—	11kV / 33kV

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Plumbing & Piping

Domestic Water, Drainage, Fire Protection & Piping Systems

Piping Systems in Revit

- Pipe (PI) command: Systems tab → Pipe → draw between equipment/fixtures
- Pipe Types: Set material (copper, PVC, carbon steel, stainless) and joining method
- System Classification: Domestic Cold/Hot, Hydronic Supply/Return, Fire Protection, Fuel
- Slope: Set pipe slope in Properties → slope value (mm/m) for gravity drainage
- Pipe Sizing: Analyse tab → Pipe Pressure Loss Report; or set size manually
- Insulation: Select pipe → Add Insulation → choose material/thickness

Plumbing Fixture Placement

- Load plumbing fixture families (WCs, basins, showers) from Revit library or manufacturer
- Place face-based fixtures on wall/floor — ensure correct rotation and offset
- Connect: Select fixture → Connect → pipe appears; complete routing back to riser
- Drainage: Model horizontal waste pipes with slope; connect to vertical stacks
- Hot/Cold: Route separate systems; colour-code by system type for coordination drawings
- Fixture Units (FU): Input in fixture properties for demand load calculation

PIPE SIZING IN REVIT:

1. Set system type and design flow rates in system properties
2. Analyse tab → Pipe Pressure Loss Report → review friction and velocity
3. Target velocity: 1.2–2.0 m/s domestic; 1.5–2.5 m/s chilled water
4. Friction loss: 200–400 Pa/m for domestic; 150–300 Pa/m for HVAC pipework
5. Manually size critical sections; use Revit sizing tool for remainder

Pipe System		Colour Code (CIBSE)	Typical Material	Slope Required
Domestic Water	Cold	Blue	CPVC / Copper	No (pressurised)
Domestic Water	Hot	Red	Copper / Stainless	No (pressurised)
Chilled Supply	Water	Cyan	Carbon steel / Cu-Ni	No
Chilled Return	Water	Cyan (dashed)	Carbon steel / Cu-Ni	No
HWS Supply		Orange	Carbon steel	No
Soil & Waste (drain)		Grey / Brown	uPVC	Yes: 1:80 minimum
Fire Sprinkler		Red (dashed)	Black steel / CPVC	Drain slope only

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Families

System Families, Loadable Components & Family Editor

Types of Revit Families

Family Type	Definition	Examples	Editable?
System Family	Built into Revit; not in library	Ducts, pipes, walls, floors	Properties only
Loadable Family	External .rfa files; loaded into project	Luminaires, FCUs, valves	Full editor access
In-Place Family	Created directly in project for unique items	Custom plenums, cable routes	Project-specific
Annotation Family	2D symbols for documentation	Tags, title blocks, North arrow	Full editor access

Family Editor — MEP Connector Setup

- CREATING AN MEP EQUIPMENT FAMILY:
1. New Family → select template: Mechanical Equipment, Electrical Equipment, etc.
 2. Create 3D geometry (Extrusion/Blend) matching equipment envelope
 3. Add MEP Connector: Create tab → Connector → place on face at connection point
 4. Set connector: System Classification (Duct Air/Pipe Hydronic/Electrical)
 5. Add parameters: Flow, Pressure, Voltage — make them instance or type params
 6. Load into Project: Family editor → Load into Project (Ctrl+Shift+O)

Managing Family Libraries

- Autodesk Content Library: Revit/Content folder — basic families included
- Manufacturer BIM Libraries: Danfoss, Grundfos, Schneider, Legrand provide Revit families
- BIMobject / NBS Source: Online repositories — verify LOD and parameter completeness
- Company Library: Create standardised, tested families for consistent project use
- Purge Unused: Manage tab → Purge Unused → remove unneeded families (reduces file size)
- Transfer Project Standards: Manage → Transfer Project Standards → between projects

	ARCH	STRUCT	MECH	ELEC	PLUMB
ARCH	-	S	H	M	M
STRUCT		-	M	H	M
MECH			-	H	H
ELEC				-	H
PLUMB					-

H = High clash risk
 M = Medium risk
 S = Single clash

Discipline Clash Risk Matrix

Discipline Clash Risk Matrix — High/Medium/Standard

Navisworks Clash Detection Workflow

1. Export: Revit → Navisworks (.nwc) — each discipline exports separately
2. Append in Navisworks: File → Append all discipline files to form federated model
3. Clash Detective: Home tab → Clash Detective → New Test
4. Select A (MECH) vs B (STRUCT/ARCH) — set tolerance (typically 25mm)
5. Run test → review results → group and assign clashes to responsible party
6. Status tracking: Assign → Active → Approved/Resolved → Reviewed
7. Report: Clash Detective → Report → export PDF/HTML for coordination meeting

Coordination View Setup in Revit

- Create 3D coordination view: Hide all linked models except discipline of interest
- Override Graphics: Colour-code each system by type (View → Override Graphics in View)
- Section Box: Use section box to isolate coordination area/zone
- Interference Check: Collaborate tab → Interference Check → between elements in project
- Revit Cloud Worksharing: BIM 360 Coordination for issue tracking and markup

Common MEP Clash Scenarios

Clash Type	Typical Cause	Resolution	Priority
Duct vs Beam	Structural depths not allowed for	Offset duct, raise beam if possible	High
Pipe vs Conduit	Route conflicts at ceiling zone	Vertical separation (pipe above)	High
Duct vs Door frame	Insufficient ceiling-to-soffit	Reduce duct size or reroute	High
Cable Tray vs Structure	Tray spans over beams	Step tray up/down or penetrate	Medium
Sprinkler vs Light	Grid ceiling coordination	Adjust layout in coordination model	Medium

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Schedules & Annotation

Equipment Schedules, Tags & Sheet Annotation

Creating MEP Schedules

- View tab → Schedules → Schedule/Quantities → select category (e.g., Mechanical Equipment)
- Fields: Add parameters you need — Family, Type, Mark, System Name, Flow, Power, etc.
- Filter: Show only specific system types or levels (Filter by "System Classification" = Supply Air)
- Sorting/Grouping: Sort by mark number; group by system or level
- Formatting: Set column width, header text, number formatting
- Key Schedule: Create lookup key for repeated parameter combos (e.g., insulation type codes)

Tag Families for MEP

TAGGING MEP ELEMENTS:

Annotate tab → Tag All Not Tagged → select category → apply to view

Common tags: Duct Tag (size+system), Pipe Tag, Space Tag, Equipment Tag

Tag Leaders: Enable leader in type properties for space-constrained annotations

Keynote: Annotate → Keynote → By Element — links to specification system

Room/Space Tags: Show ventilation zone name, number, area in RCP plans

Useful MEP Schedule Types

Schedule Name	Category	Key Parameters	Use
Equipment Schedule	Mechanical Equipment	Tag, Type, Capacity, Power	Specifications & BOQ
Duct Schedule	Ducts	Length, Width, Height, System	Quantity take-off
Panel Schedule	Electrical Equipment	Circuits, Loads, Breaker size	Electrical drawings
Pipe Schedule	Pipes	Length, Diameter, Material, System	Quantity & procurement
Space Schedule	Spaces	Area, Volume, Occupancy, Air Flow	Compliance & ventilation
Lighting Fixture	Lighting Fixtures	Mark, Watts, Lumens, Circuit	Energy analysis

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Views & Sheet Production

Plan Views, Sections, Sections & Drawing Sheets

View Management Best Practices

- View Template: Apply to all similar views — controls visibility, detail level, scale
- View Range: Set top/cut/bottom for correct plan display (especially for reflected ceiling plans)
- Discipline: Set Mechanical, Electrical, Coordination per view — controls default visibility
- Detail Level: Coarse (schematic), Medium (construction), Fine (fabrication)
- VG Overrides (VV): Override visibility per element category or filter per view
- Filters: Create rule-based filters (e.g., all Supply Air ducts → cyan colour fill)

Sheet Setup & Titleblock

SHEET NUMBERING CONVENTION (ISO/BS 1192):

M-001: Mechanical — Floor Plans

E-001: Electrical — Power Plans

P-001: Plumbing — Domestic Water Plans

FP-001: Fire Protection — Sprinkler Plans

C-001: Coordination — Combined MEP Coordination Plans

SC-001: Schematics / Riser Diagrams per discipline

Commonly Required MEP Drawings

Drawing Type	Scale	Contents	View Type
Floor Plan — HVAC	1:100	Ductwork, equipment, diffusers, space names	Plan (RCP)
Floor Plan — Electrical	1:100	Conduit, panels, outlets, lighting	Plan (ceiling/floor)
Floor Plan — Plumbing	1:100	Pipe runs, fixtures, risers	Plan
Schematic/Riser Diagram	NTS	System schematic, connectivity, sizing	Drafting view
Equipment Room Plan	1:50	AHU, chillers, pumps with clearances	Plan
Coordination Plan	1:100	All services combined (clash check)	3D / plan coordination
Detail Sheet	1:20/ 1:10	Penetration, support, connection detail	Detail / section

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IFC & Data Export

Open BIM, COBie, IFC Export & Interoperability

IFC Export from Revit

- File → Export → IFC → use IFC Export Setup (configure before first export)
- IFC Version: Use IFC4 for new projects; IFC2x3 for legacy software compatibility
- Level of Detail: Set matching project LOD (100–400) for intended use
- Property Sets: Map Revit parameters to IFC property sets (use IFC Export Settings)
- Coordinate: Export with shared coordinates (not project internal) for federation
- Element Mapping: Verify IFC entity mapping (IFCDuct, IFCPipeSegment, IFCLightFixture)

IFC ENTITY MAPPING — KEY MEP TYPES:

Duct → IFCDuctSegment / IFCDuctFitting

Pipe → IFCPipeSegment / IFCPipeFitting

Air Terminal → IFCAirTerminal

AHU → IFCUnitaryEquipment

Chiller → IFCChiller

Electrical Panel → IFCElectricDistributionBoard

Cable Tray → IFCCableCarrierSegment

COBie Data for Facilities Management

- COBie (Construction Operations Building Information Exchange) — structured asset data for FM
- Revit COBie Extension: Export via Manage → COBie Export → Excel/IFC-based output
- Key COBie sheets: Facility, Floor, Space, Zone, Component, System, Attribute
- Component: Each maintainable asset (AHU, fan, valve) = one COBie Component row
- Attribute: Additional data (serial number, warranty, maintenance interval, manufacturer)
- Required at project completion for asset handover to FM team per ISO 19650 Part 3

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Dynamo & API

Revit API, Dynamo Scripting & Automation

Dynamo for MEP Automation

Dynamo is a visual programming environment integrated with Revit. MEP engineers use Dynamo to automate repetitive tasks, generate elements from external data, and perform complex geometric operations that would be impractical through the Revit UI.

- Access: Manage tab → Visual Programming → Dynamo Player / Dynamo (editor)
- Common MEP uses: Auto-number equipment tags, populate schedules from Excel, place outlets on a grid
- Nodes: Revit.Elements.Element → Get/SetParameterByName for batch parameter changes
- Data-Shapes: Package for user input forms in Dynamo scripts (install via Package Manager)
- Select All Elements of Category: Get all ducts/pipes on level → filter → set parameter

Common Dynamo MEP Scripts

Script Purpose	Input	Output	Complexity
Auto-number Duct Tags	Duct selection, prefix string	Sequential tag marks	Low
Space to Zone Mapping	Space schedule CSV	Zone assignments in model	Medium
Equipment from Excel	Excel table (mark, type, loc)	Placed families with params	High
Parameter Copy	Source parameter, target	Mapped data across elements	Low
Duct Length Report	Level/system filter	CSV with lengths per system	Medium
Circuit Load Balancing	Panel + circuit data	Suggest phase assignment	High

REVIT API (Python/C#):

FilteredElementCollector: Query all elements of a category in a document

Transaction: All model changes must be wrapped in a Transaction context

pyRevit: Python-based Revit macro framework — install for rapid script development

Use cases: Batch export, custom validation, BIM health checks, automated QA reports

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Best Practices & Careers

Standards, Quality Control & MEP BIM Career Paths

MEP BIM Standards

- ISO 19650 Parts 1–5: International BIM standard — information management over asset lifecycle
- BS EN ISO 19650-2: Delivery phase — EIR, BEP, OIR, AIR document structure
- RIBA Plan of Work 2020: UK project stages (0–7) with BIM responsibility matrix
- AIA E203 / G202 (US): BIM protocol document and Project BIM Plan
- COBie UK 2012 / COBie Schema 2.4: Asset data handover standard
- IFC4 ADD2: OpenBIM interoperability standard for all disciplines

Model Quality Control Checklist

- PRE-ISSUE MODEL QC:
1. All elements on correct worksets and correct level
 2. No unconnected duct/pipe ends (use Manage → Warnings)
 3. All spaces enclosed and tagged (no open boundary warnings)
 4. Equipment marks are unique and follow project convention
 5. All systems have equipment assigned (no orphan terminals)
 6. Purge unused families, types, materials before issue
 7. Run Interference Check — resolve or document accepted clashes

Revit MEP Career Pathways

Role	Focus	Tools	Salary Range
BIM Technician / Drafter	Model production, drawing output	Revit, Navisworks	\$45K–65K
MEP BIM Engineer	System design + modelling	Revit, Hevacomp/EnergyPlus	\$60K–85K
BIM Coordinator	Multi-discipline coordination	Revit, Navisworks, BIM360	\$70K–95K
BIM Manager	Standards, templates, QC	Revit, Dynamo, Forge API	\$90K–120K
Digital Engineering Lead	Digital delivery strategy	All + cloud platforms	\$110K–150K+

- Certifications: Autodesk Certified Professional (ACP) — Revit MEP; buildingSMART Professional
- BSc/MSc in Mechanical/Electrical Engineering is baseline; BIM is the differentiator
- Portfolio: Showcase coordinated federated models, Dynamo scripts, clash resolution reports